APPENDIX I

Burkina Faso—Joint Fund-World Bank Debt Sustainability Analysis

1. **This joint DSA was prepared using the Fund-World Bank debt sustainability framework for low-income countries.** The data were updated jointly by the Fund and the World Bank with information provided by the authorities. The medium-term framework was discussed with the authorities in the context of the new PRGF arrangement (Box I.1), which is being considered by the Fund Board at the same time as this DSA.

2. **Burkina Faso's risk of debt distress is moderate.**

A. External Debt Sustainability Analysis

3. **All debt indicators are well below their policy-dependent thresholds in the baseline scenarios** (Text Table I.1). This is a substantial improvement over the results of the previous joint DSA completed in August 2005, when the NPV of debt-to-exports ratio was projected to breach its threshold for a number of years (Box I.2).[10] After the MDRI debt relief in 2006, debt ratios increase again over the medium term and stabilize over the long term. The driver of these debt dynamics is the fiscal stance; over the medium term, fiscal deficits are relatively large and the borrowing requirements lead to increasing debt ratios, but in the long run deficits fall enough to allow debt ratios to stabilize, mainly because of projected revenue increases.

Text Table I.1: Policy-Based External Debt Burden Indicators

	Threshold [1]	Burkina Faso's ratios		
		2006	2007-25 [2]	Peak
NPV of external debt in percent of:				
Exports	200	85.8	152.5	164.5
GDP	50	10.6	19.3	21.4
Revenues	300	85.4	113.6	120.5
External debt service in percent of:				
Exports	25	5.5	6.7	8.8
Revenues	35	5.5	4.9	6.2

[1] Policy-dependent thresholds as used in the joint IMF-World Bank LIC DSA framework for a strong policy performance. Burkina Faso received a rating of 3.76 in the World Bank's 2005 Country Policy and Institutional Assessment, which qualifes it as a strong performer. The staffs decided against using the three-year moving average CPIA score to avoid undue volatility in the applicable thresholds.
[2] Simple average.

[10] See IMF Country Report No. 05/354, September 30, 2005.

Box I.1. Burkina Faso—Macroeconomic Assumptions

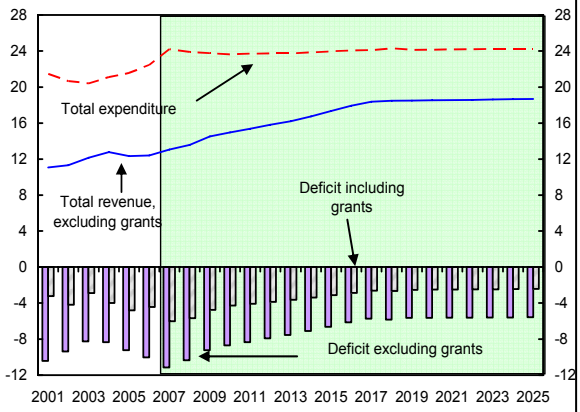
Real GDP growth is projected to average 6 percent through 2025, in line with the historical average of the past ten years. This is consistent with annual per capita GDP growth of 3.6 percent, which would lead to a doubling of real per capita GDP over this period.

Inflation is projected at 2 percent–in line with the 10-year historical average and WAEMU convergence criteria.

The **overall fiscal deficit**, with and without grants, is projected to shrink over time as revenue increases. Strengthening tax and customs administration and reforming the tax system is at the center of the proposed new PRGF program; over the long run, revenue projections are consistent with the authorities' commitment to reach the WAEMU tax revenue target of 17 percent of GDP. Expenditures are programmed to expand considerably in 2007 relative to GDP and stay at the higher level throughout the projection period. External financing requirements are to be met equally by grants and loans, in line with recent experience. The decline in the overall deficit including grants translates into a broadly equal reduction in external new borrowing, because domestic financing is assumed to be zero for most years because its costs are high.

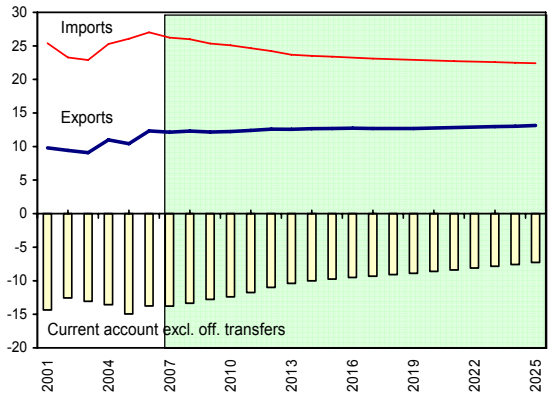
The **current account deficit** is expected to narrow moderately through 2025 as the tighter fiscal stance dampens import demand. The recent increase in the import-to-GDP ratio caused by higher oil prices is likely to be partially reversed because prices are expected to be lower, and the connection of the transmission grid with Côte d'Ivoire should reduce the consumption of oil for electricity generation. The exports-to-GDP ratio is projected to increase slightly over the projection period. The recent increase was brought about by strong cotton exports; future export growth would come from other sources.

Fiscal Developments, 2001–2025
(In percent of GDP)



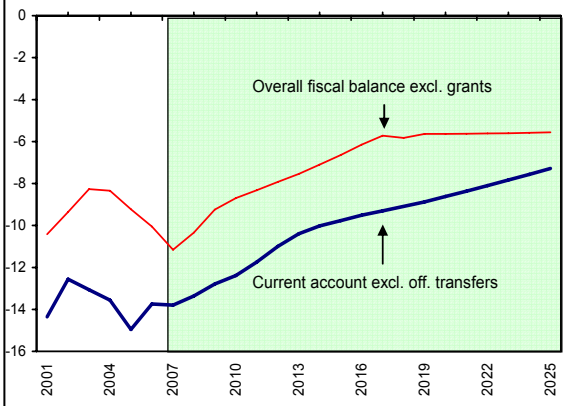
Source: Burkinabè authorities and staff estimates.

Current Account, Exports, and Imports 2001-25
(In percent of GDP)



Source: Burkinabè authorities and staff estimates.

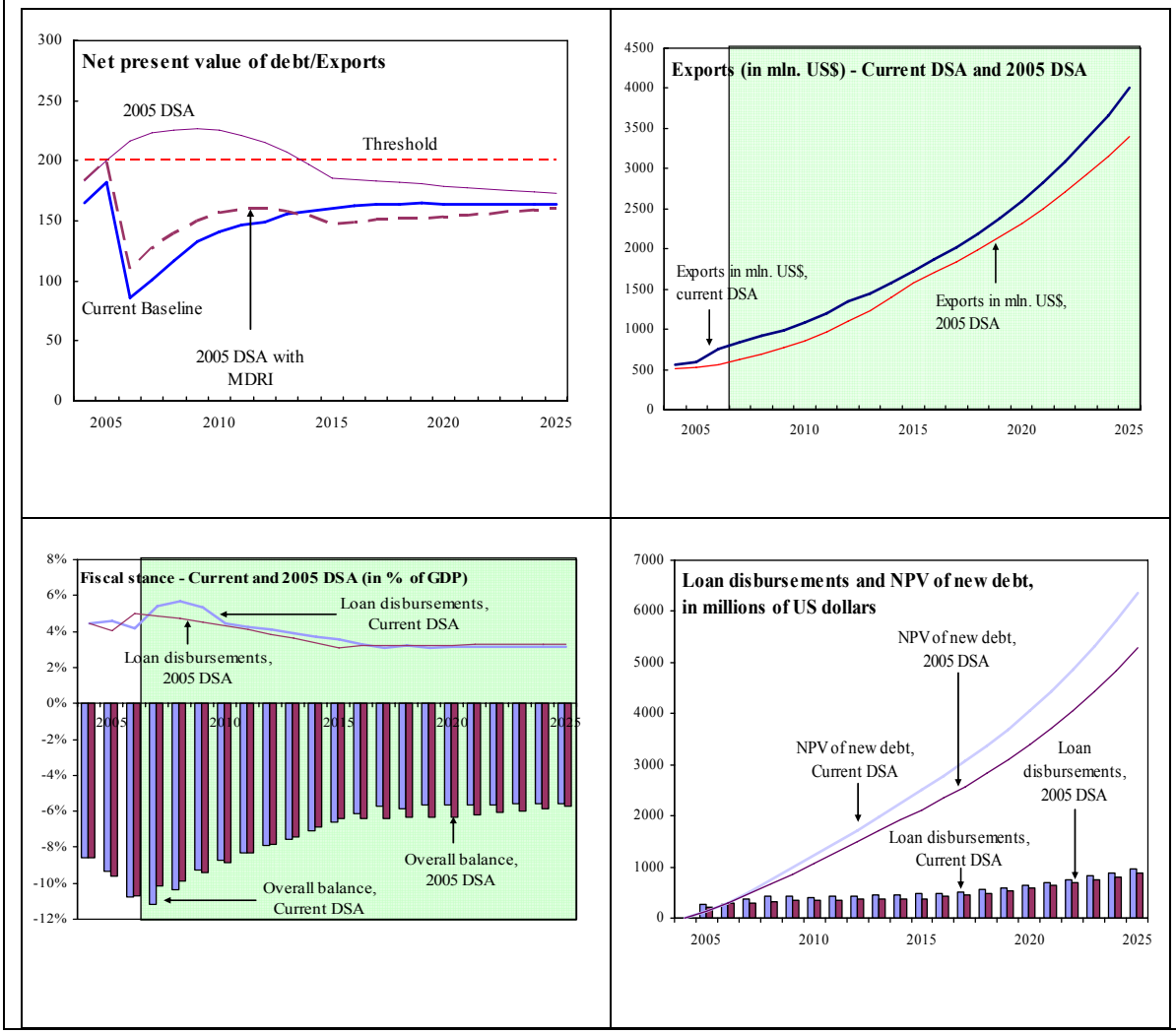
Fiscal and Current Account Balances 2001-25
(In percent of GDP)



Source: Burkinabè authorities and staff estimates.

Box I.2. Burkina Faso—Changes From the 2005 Joint DSA

- **MDRI relief**—in the first figure, MDRI relief has been simulated for the 2005 joint DSA, yielding the dotted line, which tracks the current baseline.
- **More favorable macroeconomic performance and outlook**—this is the difference between the current baseline and the 2005 DSA with MDRI over the medium term: unexpectedly good export performance in recent years (second figure), particularly in 2006, creates the basis for a more favorable macroeconomic outlook.
- **Larger new borrowing requirements in nominal terms**—this tends to offset the more favorable export path in the baseline, and the debt ratios of the two DSAs converge (see third figure below). As a percent of GDP, projections for the fiscal stance—as measured by the overall balance excluding grants—have changed relatively little since 2005, except for 2007-09, when the current baseline has larger deficits. In U.S. dollar terms, borrowing requirements in the current baseline are somewhat larger in most years, because nominal GDP is larger. While the differences in individual years tend to be small, they accumulate over time, and the NPV of new debt diverges (fourth figure).



4. **The NPV of debt-to-exports ratio is the most binding indicator for Burkina Faso.** Sensitivity tests show that the ratio breaches its threshold in a number of scenarios, but other debt indicators remain below their thresholds for all standardized stress tests and alternative scenarios (Text Table I.2 and Figure I.1):

- The **stress test on export growth** causes the NPV of debt-to-exports ratio to breach its threshold by a wide margin because exports are volatile: they are heavily dependent on cotton and there are large fluctuations caused by weather-related shocks and changes in international cotton prices.[11]
- **Public borrowing on less favorable terms** would probably be unsustainable. In the alternative scenario, the average grant element is lowered to about 20 percent from over 40 percent in the baseline (which increases the interest rate on new borrowing by 2 percent), resulting in a steadily increasing NPV of debt-to-exports ratio, which breaches the threshold in about 2015.
- In the **alternative historical scenario**, the NPV of debt-to-exports ratio also breaches its threshold. In this scenario, the growth rates for key macroeconomic variables are set at their historical 10-year average. Debt dynamics in Burkina Faso are mostly driven by fiscal policies—the country can draw on the regional reserves of the WAEMU to meet external balance-of-payment needs—and past policies appear to be unsustainable. In contrast, the baseline assumes a significant reduction in the fiscal deficit over the medium term, which is critical for debt sustainability.

5. **A country-specific scenario that retains the fiscal policy stance of the last year of the PRGF indefinitely would also be unsustainable.** In this scenario, the revenue-to-GDP ratio is fixed at its projected level for 2009 (14.5 percent of GDP), but baseline assumptions for expenditures excluding interest are broadly unchanged (Figure I.2).[12] The overall deficit including grants would then average about 4½ percent of GDP for 2010–25, about 1½ percentage points higher than the baseline. The additional new borrowing requirements would lead to a continuous increase in the NPV of debt-to-exports ratio, eventually breaching the threshold in about 2018. This underlines the importance of reducing the deficit over the long run for safeguarding debt sustainability. In the baseline, the increase in domestic revenues reduces the deficit while keeping expenditures high.

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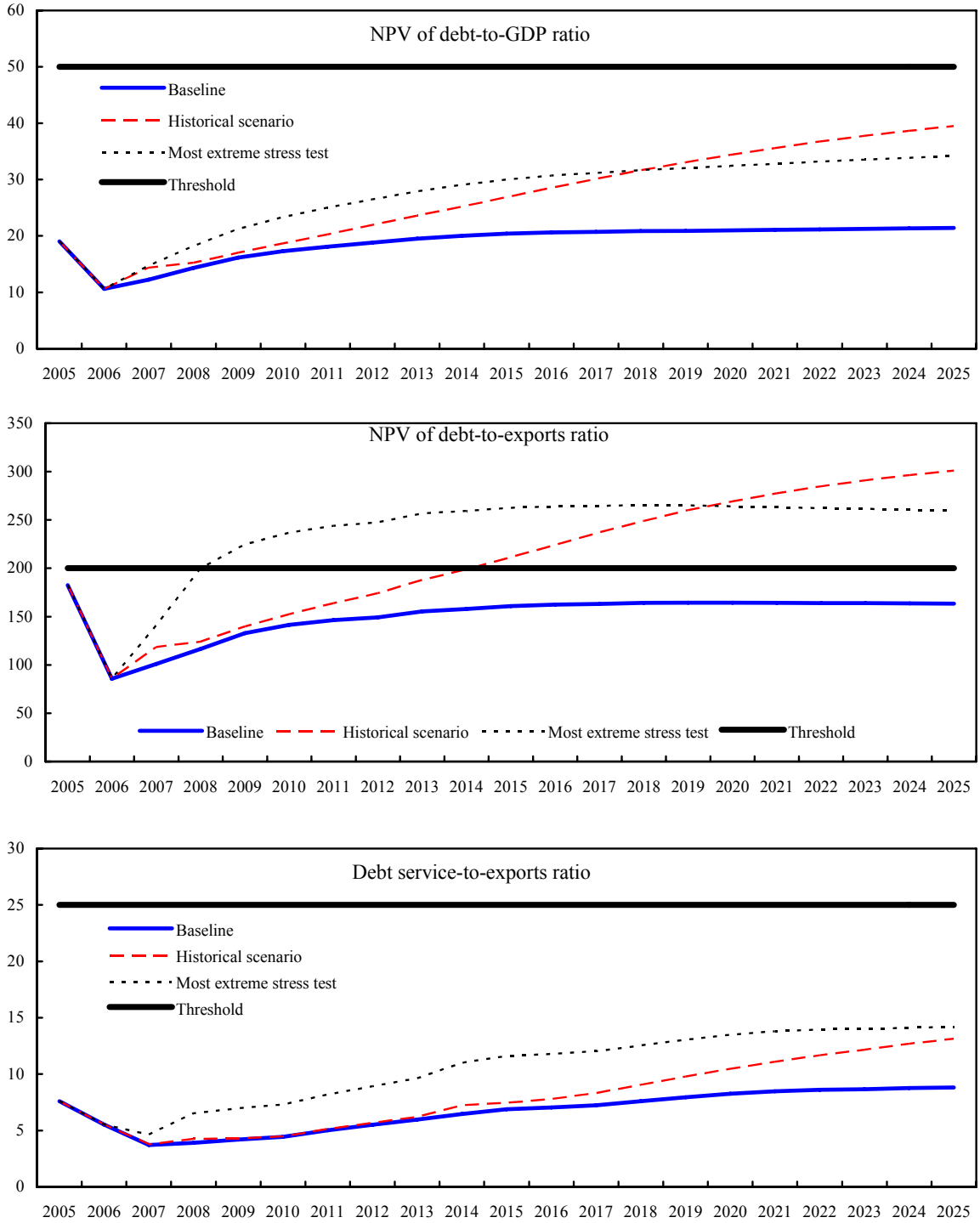
[11] The combination stress test also leads to a breach of the threshold for the NPV of debt-to-exports ratio, again on account of the large export volatility.

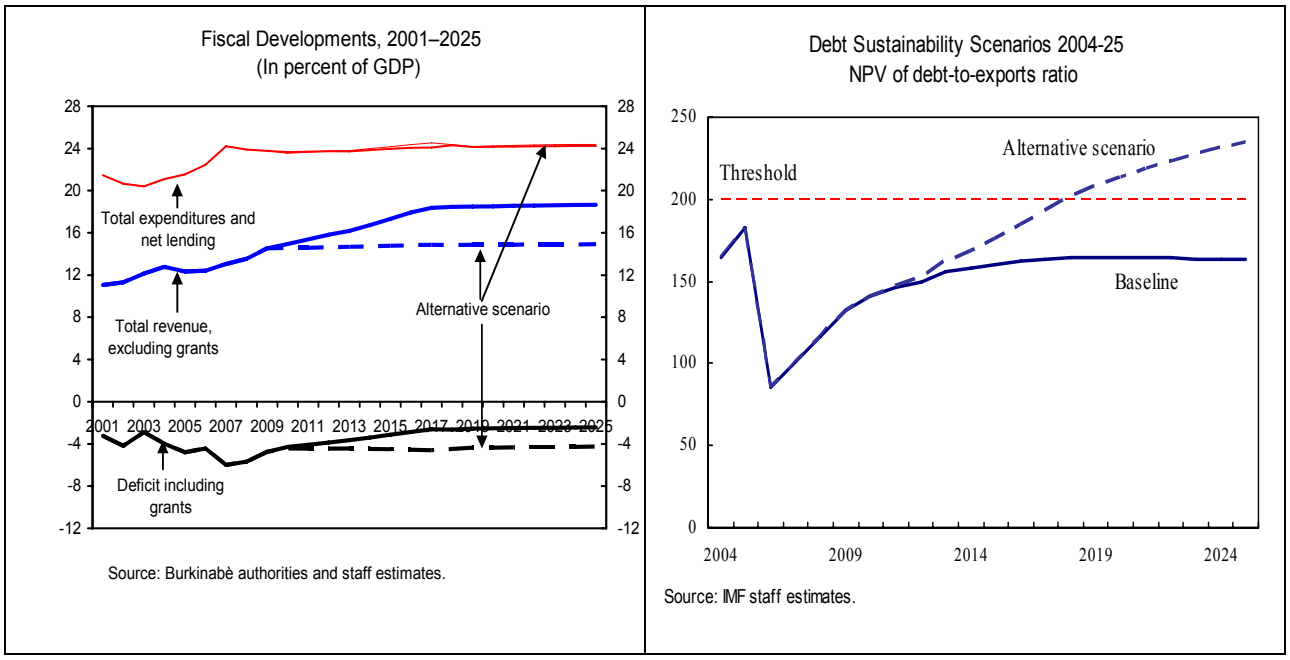
[12] Expenditures increase relative to the baseline because of the larger interest payments; real GDP growth is also higher in the alternative scenario, because growth in the baseline is dampened by the stronger revenue effort.

Figure I.1. Burkina Faso: Indicators of Public External Debt
Under Alternative Scenarios, 2005-25
(In percent)



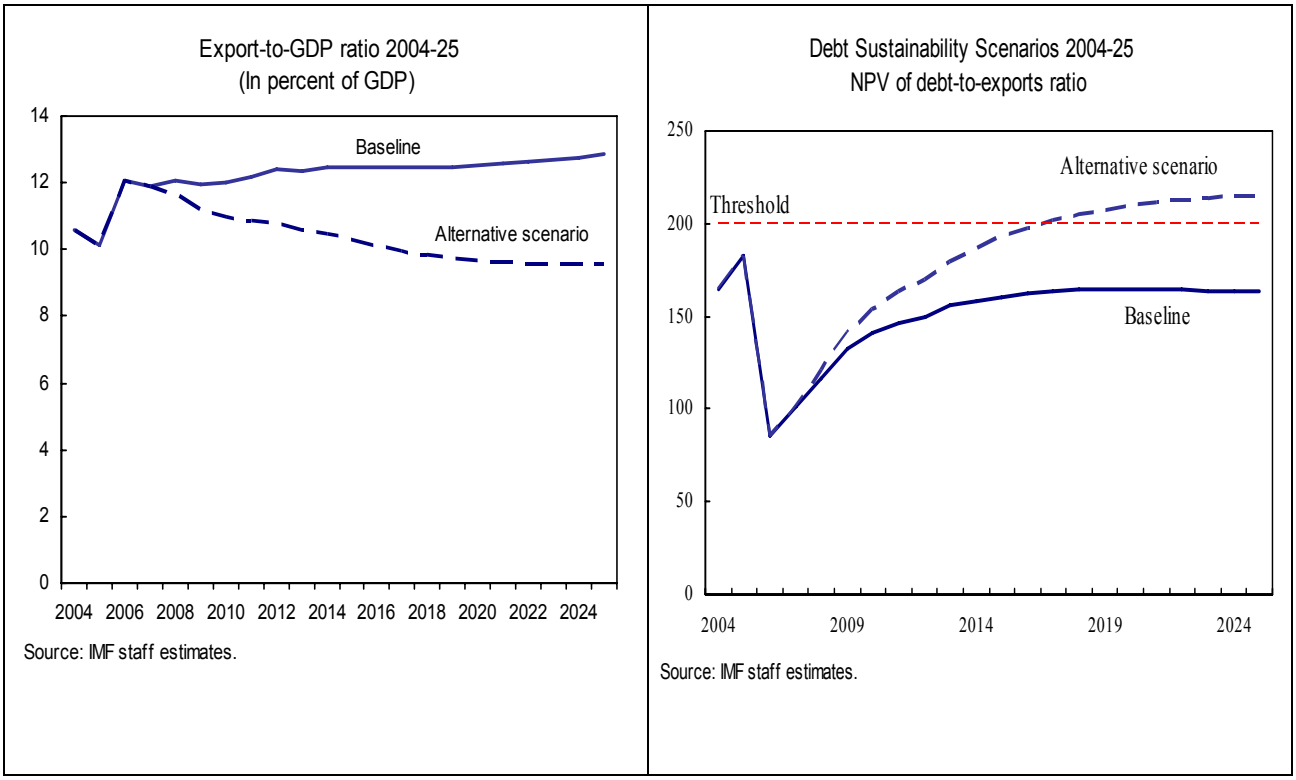
Source: IMF staff projections and simulations.

Figure I.2: Country-Specific Scenario—Less Long-Term Fiscal Consolidation



6. **A second country-specific scenario with a more pessimistic export outlook results in a marginal breach of the threshold.** Cotton exports have risen sharply in recent years, mostly because the area under cultivation is larger. The baseline assumes a slowdown in cotton export growth as the sector matures, with most output gains coming from productivity improvements. The alternative scenario assumes no productivity gains and a constant area under cultivation, so the volume of cotton exports stays at the 2007 level. As a consequence, the exports-to-GDP ratio would decline substantially relative to the baseline (Figure I.3). This would lead the NPV of debt-to-exports ratio to breach its threshold, but only marginally. Of course, that scenario would likely require additional external adjustment, which has not been considered in this analysis.

Figure I.3: Country-Specific Scenario—Weaker Export Growth

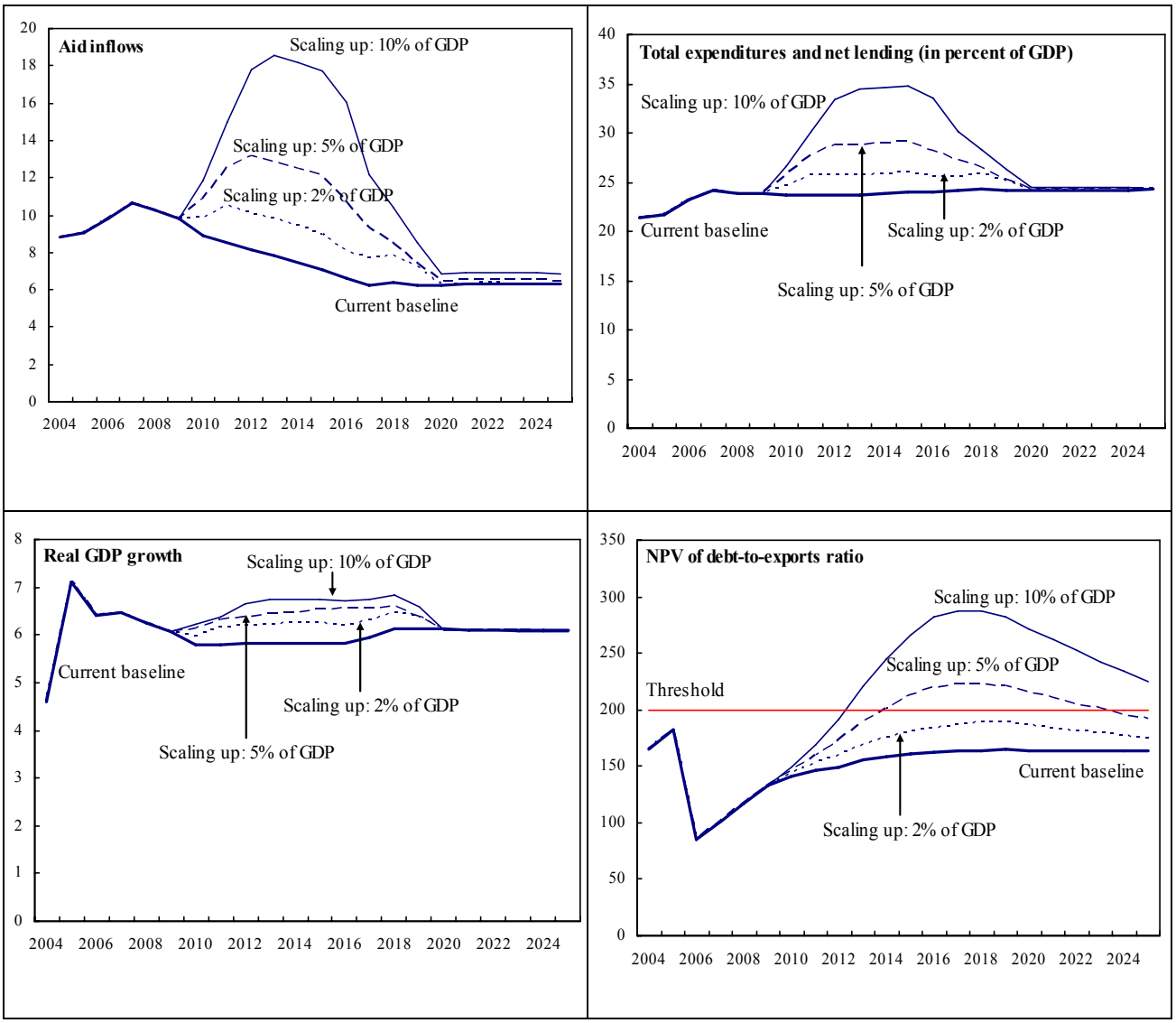


7. **A country-specific scaling-up of aid scenario makes it clear that additional aid should be in the form of grants.** It is assumed that aid is scaled up between 2010 and 2015 and then scaled down to converge with the baseline by 2020. Three scenarios are considered: (i) a small scaling-up with aid inflows increasing by 2 percent of GDP by 2015, equivalent to an aid increase of about 20 percent relative to the average aid-to-GDP ratio in 2007–09; (ii) a moderate scaling-up with aid inflows of 5 percent of GDP, an increase of about 50 percent; and (iii) a large scaling-up with aid of 10 percent of GDP, i.e., a doubling of aid (Figure I.4). The impact on growth is modeled on cross-country empirical evidence, taking into account the declining marginal returns of aid.[13] As in the baseline, a 50:50 split between loans and grants is assumed. In spite of the high grant share and borrowing on highly concessional terms, the NPV of debt-to-exports ratio increases noticeably in all three scenarios and

[13] See Michael Clemens, Steven Radelet, and Rikhil Bhavani (2004), "Counting Chickens When They Hatch: The Short Term Effect of Aid on Growth", Center for Global Development, Working Paper No. 44. The authors distinguish between different types of aid and focus in their growth regressions on "so-called short-impact" aid, which is most likely to have a growth impact over the medium term. Simulations are based on the authors' preferred conservative specification, which is that short-impact aid equivalent to 1 percent of GDP has an impact on growth of about 0.5 percent if the country receives no aid. Given the already large aid inflows under the baseline, a 1 percent of GDP increase in aid would raise growth about 0.2 percent, declining further to about 0.1 percent under the large scaling-up scenario.

breaches its threshold in the moderate and large scenario. This suggests that a scaling up of aid should rely mostly on grants to safeguard debt sustainability.

Figure I.4: Country-Specific Scenario—Scaling-Up Aid



B. Public Debt Sustainability Analysis

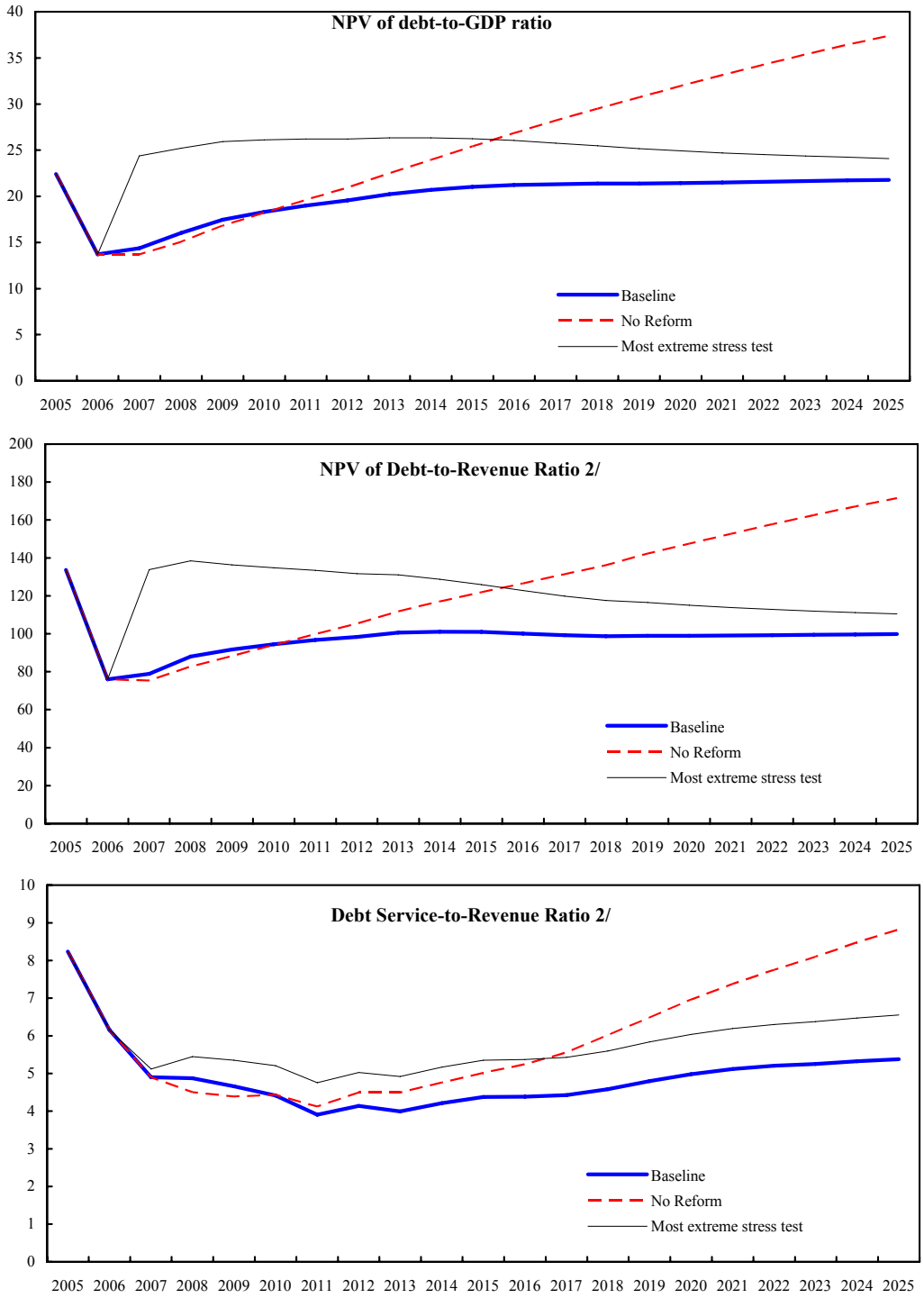
8. **The domestic debt stock in Burkina Faso is low.** As of September 2006 domestic debt totaled less than 3 percent of GDP. Following the large inflow of privatization receipts in January 2007, the government is practically free of domestic debt. The fiscal framework envisions zero domestic financing for most of the projection period, keeping domestic debt very low. Thus, public debt dynamics are similar to those of external debt.

9. **The public debt "no reform" alternative scenario confirms that present deficits are too high to stabilize debt ratios.** The 'no reform' scenario (Table I.4 and Figure I.5) freezes the primary deficit at its 2006 value—about 4¾ percent of GDP—and simulates the resulting debt dynamics. This yields continuously increasing debt ratios—this is not surprising because the 2006 primary deficit is higher than that necessary to stabilize the nominal debt-to-GDP ratio at its post-MDRI level (Box I.3).

C. Conclusions

10. **Burkina Faso's risk of debt distress has moderated.** While all baseline scenarios are comfortably below their policy-dependent thresholds, they depend critically on the assumption that domestic revenues continue to increase and lead to declining deficits after grants, thus reducing new borrowing requirements. The risk embodied in domestic revenue performance is compounded by Burkina's vulnerable export performance, which is essentially based on one product, cotton. MDRI relief has reduced the risk of debt distress, but risks pertaining to revenue and export performance remain. Thus, it is essential to implement the revenue reform program forcefully and to work on broadening the export base.

Figure I.5. Burkina Faso: Indicators of Public Debt
Under Alternative Scenarios, 2005-25 1/
(In percent)



Source: Staff projections and simulations.
1/ Most extreme stress test is test that yields highest ratio in 2015.
2/ Revenue including grants.

Box I.3. Burkina Faso—Standard Debt Dynamics

The link between the fiscal stance and debt dynamics in Burkina Faso can be considered in a standard model where the evolution of the public debt-to-GDP ratio ($d(t)$) is a function of the growth rate of real GDP (g), the real interest rate on government debt (r), and the primary balance on the government budget (pb):

(1) $$d(t+1) = d(t)\left[(1+r)/(1+g)\right] - pb .$$

The primary surplus needed to keep the debt ratio stable (pb^*) is:

(2) $$pb^* = \left[(r-g)/(1+g)\right] d(t) .$$

Baseline projections for Burkina Faso assume an average real GDP growth rate of about $g = 6.1\%$, and the average real interest rate is approximately $r = -0.6\%$, reflecting a very high share of external debt with very low nominal interest rates, which implies a negative real interest rate overall.

The model allows for calculating the primary balance that is consistent with stabilizing either the pre- or post-MDRI debt-to-GDP ratio (Box Table 1). The baseline scenario assumes an average primary deficit of about 1.8 percent of GDP, which is close to the debt-stabilizing primary deficit for the post-MDRI level. Hence, debt dynamics are relatively stable in the baseline. However, the debt-stabilizing deficits for pre- or post-MDRI debt ratios are much smaller than what has been observed historically—the average for 2001-06 was 3.8 percent of GDP—so alternative historical or "no reform" scenarios show increasing debt ratios throughout the projection horizon, because it takes considerable time for these scenarios, starting from low post-MDRI debt levels in 2006, to converge to their equilibrium with much higher debt levels (Box Table 2).

Box Table 1: Debt-stabilizing primary balance

Debt-to-GDP ratio	Debt-stabilizing primary balance
Pre-MDRI level: 1/ 43%	-2.8%
Post-MDRI level: 2/ 20%	-1.3%

1/ Average of public debt-to-GDP ratio for 2003-05.
2/ End 2006.

Box Table 2: Debt-to-GDP ratios for alternative primary balances

Primary balance	Simulated path for debt-to-GDP ratio 1/		
	2006	2025	2050
Average 2001-06: -3.8%	20%	49%	58%
Average 2007-09: 2/ -5.1%	20%	63%	77%
Average baseline: 3/ -1.8%	20%	26%	28%

1/ Maintains indicated primary balance throughout the projection horizon, beginning in 2007.
2/ Projection for program period.
3/ Projection for 2007-25.

Appendix Table I.1. Burkina Faso: External Debt Sustainability Framework Including Impact of MDRI, Baseline Scenario, 2004-25
(Percent of GDP, unless otherwise indicated)

	Actual	Historical Average [1]	Standard Deviation [1]	Estimate					Projections					
	2004			2005	2006	2007	2008	2009	2010	2005-10 Average	2015	2020	2025	2011-25 Average
External debt (nominal) [2]	**41.2**			**37.6**	**17.5**	**20.7**	**24.4**	**27.5**	**29.5**	**26.2**	**34.2**	**34.0**	**33.1**	33.4
Change in external debt	-0.7			-3.6	-20.2	3.2	3.7	3.1	2.0	-1.9	0.5	-0.2	-0.2	0.2
Identified net debt-creating flows	3.3			8.7	8.1	4.2	9.0	6.9	5.8	7.1	3.5	2.5	2.4	3.2
Noninterest current account deficit	**10.3**	**8.9**	**3.2**	**11.5**	**10.5**	**10.5**	**10.3**	**9.8**	**9.6**	10.4	**7.4**	**6.6**	**5.3**	6.9
Deficit in balance of goods and services	14.3			15.7	14.7	14.1	13.7	13.2	12.8	14.0	10.7	10.0	9.3	10.4
Exports	11.0			10.4	12.3	12.1	12.3	12.2	12.2	11.9	12.7	12.8	13.1	12.8
Imports	25.3			26.1	27.0	26.2	26.0	25.3	25.1	26.0	23.4	22.8	22.4	23.2
Net current transfers (negative = inflow)	-3.7	-5.8	2.6	-3.9	-4.0	-3.5	-3.2	-3.1	-2.9	-3.4	-2.9	-3.1	-3.5	-3.0
Other current account flows (negative = net inflow)	-0.3			-0.3	-0.2	-0.2	-0.2	-0.3	-0.3	-0.3	-0.4	-0.4	-0.4	-0.4
Net FDI (negative = inflow)	**-0.5**	**-0.5**	**0.2**	**-0.4**	**-0.4**	**-5.4**	**-0.4**	**-1.7**	**-2.6**	-1.8	**-2.5**	**-2.6**	**-1.5**	-2.3
Endogenous debt dynamics [3]	**-6.6**			**-2.4**	**-2.0**	**-0.8**	**-1.0**	**-1.1**	**-1.1**	-1.4	**-1.4**	**-1.5**	**-1.4**	-1.4
Denominator: 1+g+r+gr	1.2			1.1	1.1	1.1	1.1	1.1	1.1	1.1	1.1	1.1	1.1	1.1
Contribution from nominal interest rate	0.3			0.3	0.2	0.2	0.2	0.3	0.3	0.3	0.4	0.4	0.4	0.4
Contribution from real GDP growth	-1.6			-2.7	-2.2	-1.0	-1.2	-1.4	-1.5	-1.6	-1.8	-1.9	-1.9	-1.8
Contribution from price and exchange rate changes	-5.2			…	…	…	…	…	…	...	…	…	…	
Residual (3-4) [4]	**-4.0**			**-12.3**	**-28.3**	**-1.0**	**-5.3**	**-3.8**	**-3.8**	-9.1	**-3.0**	**-2.7**	**-2.6**	-3.0
Of which: Exceptional financing	0.0			0.0	-21.5	0.0	0.0	0.0	0.0	-3.6	0.0	0.0	0.0	0.0
NPV of external debt [5]	18.1			19.0	10.6	12.2	14.3	16.2	17.3	14.9	20.4	21.0	21.4	20.5
In percent of exports	164.4			182.5	85.8	101.0	116.6	132.8	141.2	126.7	160.6	164.3	163.2	160.4
Debt service-to-exports ratio (percent)	8.1			7.6	5.5	3.7	3.9	4.2	4.4	4.9	6.9	8.3	8.8	7.4
Total gross financing need (millions of U.S. dollars)	550.3			670.1	654.9	375.0	778.2	694.8	665.3	639.7	782.0	1,029.1	1,526.6	984.8
Noninterest current account deficit that stabilizes debt ratio	11.0			15.1	30.6	7.3	6.6	6.6	7.6	12.3	6.9	6.7	5.5	6.7
Key macroeconomic assumptions														
Real GDP growth (in percent)	4.6	6.3	2.1	7.1	6.4	6.5	6.3	6.1	5.8	6.3	5.8	6.1	6.1	6.0
GDP deflator in US dollar terms (change in percent)	14.3	4.3	11.8	2.7	1.2	6.2	2.7	2.8	3.0	3.1	2.3	2.2	2.2	2.5
Effective interest rate (percent) [6]	0.8	0.4	0.4	0.7	0.7	1.2	1.2	1.3	1.3	1.0	1.3	1.3	1.4	1.3
Growth of exports of G&S (U.S. dollar terms, percent)	44.8	10.3	21.7	4.2	27.6	11.0	10.8	7.9	9.6	11.8	8.5	9.0	9.1	9.1
Growth of imports of G&S (U.S. dollar terms, percent)	32.2	11.0	15.3	13.3	11.6	9.7	8.3	6.2	7.9	9.5	7.7	8.1	8.1	7.8
Grant element of new public sector borrowing (percent)	...	...	...	46.3	44.9	44.5	43.9	44.1	45.6	44.9	44.0	41.1	38.6	42.3
Memorandum item:														
Nominal GDP (millions of U.S. dollars)	5,114			5,624	6,055	6,844	7,471	8,145	8,878		13,537	20,315	30,509	

Source: Burkinabè authorities; and IMF staff estimates and projections.

[1] Historical averages and standard deviations are generally derived over the past 10 years, subject to data availability.

[2] Includes public external debt.

[3] Derived as [r - g - r(1+g)]/(1+g+r+gr) times previous period debt ratio, with r = nominal interest rate; g = real GDP growth rate, and r = growth rate of GDP deflator in U.S. dollar terms.

[4] Includes exceptional financing (i.e., changes in arrears and debt relief); changes in gross foreign assets; and valuation adjustments. For projections also includes contribution from price and exchange rate changes. Includes also capital transfers—in particular project grants, which are projected to average over 3 percent of GDP in 2007-10—and private, non-debt creating capital inflows.

[5] Assumes that NPV of private sector debt is equivalent to its face value.

[6] Current-year interest payments devided by previous period debt stock.

Appendix Table I.2. Burkina Faso: Sensitivity Analysis for Key Indicators of Public External Debt, 2005-25
(In percent)

	Est.	Projections						
	2005	2006	2007	2008	2009	2010	2015	2025
NPV of debt-to-GDP ratio								
Baseline	19.0	10.6	12.2	14.3	16.2	17.3	20.4	21.4
A. Alternative Scenarios								
A1. Key variables at their historical averages in 2007-25 [1]	19.0	10.6	14.4	15.3	17.0	18.7	26.9	39.5
A2. New public sector loans on less favorable terms in 2007-25 [2]	19.0	10.6	14.7	18.2	21.2	23.3	30.0	34.2
B. Bound Tests								
B1. Real GDP growth at historical average minus one standard deviation in 2007-08	19.0	10.6	12.5	14.9	16.8	18.0	21.2	22.3
B2. Export value growth at historical average minus one standard deviation in 2007-08 [3]	19.0	10.6	13.7	15.7	17.5	18.5	21.3	21.7
B3. US dollar GDP deflator at historical average minus one standard deviation in 2007-08	19.0	10.6	14.0	18.3	20.6	22.0	26.0	27.2
B4. Net non-debt creating flows at historical average minus one standard deviation in 2007-08 [4]	19.0	10.6	15.1	16.8	18.5	19.5	22.0	22.0
B5. Combination of B1-B4 using one-half standard deviation shocks	19.0	10.6	13.4	15.6	17.7	19.0	22.6	24.0
B6. One-time 30 percent nominal depreciation relative to the baseline in 2007 [5]	19.0	10.6	17.1	20.0	22.5	24.1	28.4	29.8
NPV of debt-to-exports ratio								
Baseline	182.5	85.8	101.0	116.6	132.8	141.2	160.6	163.2
A. Alternative Scenarios								
A1. Key variables at their historical averages in 2007-25 [1]	182.5	85.8	118.7	124.1	139.8	152.6	211.6	301.0
A2. New public sector loans on less favorable terms in 2007-25 [2]	182.5	85.8	121.7	148.0	174.5	190.6	236.3	260.4
B. Bound Tests								
B1. Real GDP growth at historical average minus one standard deviation in 2007-08	182.5	85.8	101.0	116.6	132.8	141.2	160.6	163.2
B2. Export value growth at historical average minus one standard deviation in 2007-08 [3]	182.5	85.8	141.2	199.9	224.6	237.0	262.6	259.4
B3. US dollar GDP deflator at historical average minus one standard deviation in 2007-08	182.5	85.8	101.0	116.6	132.8	141.2	160.6	163.2
B4. Net non-debt creating flows at historical average minus one standard deviation in 2007-08 [4]	182.5	85.8	124.5	136.5	152.0	159.4	173.2	167.5
B5. Combination of B1-B4 using one-half standard deviation shocks	182.5	85.8	114.3	140.3	160.6	171.3	196.5	201.7
B6. One-time 30 percent nominal depreciation relative to the baseline in 2007 [5]	182.5	85.8	101.0	116.6	132.8	141.2	160.6	163.2
Debt service-to-exports ratio								
Baseline	7.6	5.5	3.7	3.9	4.2	4.4	6.9	8.8
A. Alternative Scenarios								
A1. Key variables at their historical averages in 2007-25 [1]	7.6	5.5	3.8	4.3	4.3	4.5	7.5	13.1
A2. New public sector loans on less favorable terms in 2007-25 [2]	7.6	5.5	3.8	4.8	5.9	6.4	10.6	15.2
B. Bound Tests								
B1. Real GDP growth at historical average minus one standard deviation in 2007-08	7.6	5.5	3.7	3.9	4.2	4.4	6.9	8.8
B2. Export value growth at historical average minus one standard deviation in 2007-08 [3]	7.6	5.5	4.6	6.5	7.0	7.3	11.6	14.2
B3. US dollar GDP deflator at historical average minus one standard deviation in 2007-08	7.6	5.5	3.7	3.9	4.2	4.4	6.9	8.8
B4. Net non-debt creating flows at historical average minus one standard deviation in 2007-08 [4]	7.6	5.5	3.7	4.4	4.6	4.8	7.8	9.2
B5. Combination of B1-B4 using one-half standard deviation shocks	7.6	5.5	4.1	4.9	5.1	5.4	8.3	10.9
B6. One-time 30 percent nominal depreciation relative to the baseline in 2007 [5]	7.6	5.5	3.7	3.9	4.2	4.4	6.9	8.8
Memorandum item:								
Grant element assumed on residual financing (i.e., financing required above baseline) [6]	41.7	41.7	41.7	41.7	41.7	41.7	41.7	41.7

Source: IMF staff estimates and projections.

[1] Variables include real GDP growth, growth of GDP deflator (in U.S. dollar terms), non-interest current account in percent of GDP, and nondebt creating flows.

[2] Assumes that the interest rate on new borrowing is 2 percentage points higher than in the baseline, while grace and maturity periods are the same.

[3] Export values are assumed to remain permanently at the lower level, but the current account as a share of GDP is assumed to return to its baseline level after the shock (implicitly assuming an offsetting adjustment in import levels).

[4] Includes official and private transfers and FDI.

[5] Depreciation is defined as percentage decline in dollar/local currency rate, such that it never exceeds 100 percent.

[6] Applies to all stress scenarios except for A2 (less favorable financing) in which the terms on all new financing are as specified in footnote 2.

Appendix Table I.3. Burkina Faso: Public Sector Debt Sustainability Framework, Baseline Scenario, 2004-25
(Percent of GDP, unless otherwise indicated)

	Actual			Estimate						Projections			
	2004	Historical Average [1]	Standard Deviation [1]	2005	2006	2007	2008	2009	2010	2005-10 Average	2015	2025	2011-25 Average
Public sector debt [2]	43.4			42.9	19.7	22.8	26.0	28.7	30.4		34.7	33.3	
Of which: Denominated in foreign currency	38.2			39.5	16.6	20.6	24.3	27.4	29.4		34.0	33.0	
Change in public sector debt	-0.5			-0.5	-23.2	3.1	3.3	2.7	1.7		0.5	-0.2	
Identified debt-creating flows	-2.2			5.6	-22.5	4.7	3.8	2.6	1.9		0.5	-0.2	
Primary deficit	3.8	3.0	0.8	4.4	4.8	5.9	5.3	4.4	3.9	4.8	2.7	2.0	2.5
Revenue and grants	17.1			16.8	18.0	18.2	18.2	19.0	19.4		20.8	21.8	
Of which: grants	4.3			4.4	26.8	5.2	4.7	4.5	4.4		3.5	3.1	
Primary (noninterest) expenditure	20.9			21.2	22.8	24.1	23.6	23.4	23.2		23.5	23.8	
Automatic debt dynamics	-6.0			1.2	-6.2	-1.1	-1.6	-1.8	-2.0		-2.2	-2.2	
Contribution from interest rate/growth differential	-2.6			-3.6	-3.3	-1.2	-1.4	-1.6	-1.7		-2.1	-2.1	
Of which: contribution from average real interest rate	-0.7			-0.7	-0.7	0.0	-0.1	-0.1	-0.1		-0.2	-0.2	
Of which: contribution from real GDP growth	-1.9			-2.9	-2.6	-1.2	-1.3	-1.5	-1.6		-1.9	-1.9	
Contribution from real exchange rate depreciation	-3.4			4.8	-2.9	0.1	-0.2	-0.2	-0.3		...	...	
Other identified debt-creating flows	0.0			0.0	-21.1	0.0	0.0	0.0	0.0		0.0	0.0	
Privatization receipts (negative)	0.0			0.0	0.0	0.0	0.0	0.0	0.0		0.0	0.0	
Recognition of implicit or contingent liabilities	0.0			0.0	0.0	0.0	0.0	0.0	0.0		0.0	0.0	
Debt relief (HIPC and other)	0.0			0.0	-21.1	0.0	0.0	0.0	0.0		0.0	0.0	
Other (specify, e.g. bank recapitalization)	0.0			0.0	0.0	0.0	0.0	0.0	0.0		0.0	0.0	
Residual, including asset changes	1.7			-6.1	-0.7	-1.7	-0.5	0.1	-0.2		0.0	0.0	
NPV of public sector debt, *of which:*	23.3			22.4	13.7	14.4	16.0	17.5	18.3		21.0	21.8	
Denominated in foreign currency	18.1			19.0	10.6	12.2	14.3	16.2	17.3		20.4	21.4	
External	18.1			19.0	10.6	12.2	14.3	16.2	17.3		20.4	21.4	
NPV of contingent liabilities (not included in public sector debt)	0.0			0.0	0.0	0.0	0.0	0.0	0.0		0.0	0.0	
Gross financing need [3]	5.3			5.8	5.9	6.8	6.2	5.3	4.7		3.6	3.1	
NPV of public sector debt-to-revenue ratio (in percent) [4]	136.1			133.7	76.0	79.0	88.0	91.8	94.5		100.9	99.8	
Of which: External	105.6			113.4	58.8	67.2	78.8	85.0	89.3		97.9	98.2	
Debt service-to-revenue ratio (in percent) [4, 5]	8.7			8.2	6.2	4.9	4.9	4.7	4.4		4.4	5.4	
Primary deficit that stabilizes the debt-to-GDP ratio	4.3			4.9	28.0	2.8	2.1	1.7	2.2		2.2	2.2	
Key macroeconomic and fiscal assumptions													
Real GDP growth (in percent)	4.6	6.3	2.1	7.1	6.4	6.5	6.3	6.1	5.8	6.3	5.8	6.1	6.0
Average nominal interest rate on forex debt (in percent)	0.8	0.7	0.1	0.7	0.6	1.2	1.2	1.3	1.3	1.1	1.3	1.4	1.3
Average real interest rate on domestic currency debt (in percent)	0.8	3.4	3.7	2.1	4.6	1.5	3.1	3.8	4.5	3.3	3.6	2.6	3.6
Real exchange rate depreciation (in percent, + indicates depreciation)	-9.3	-1.5	12.4	13.7	...	...	...	...	...	...	...	...	...
Inflation rate (GDP deflator, in percent)	3.9	3.4	3.4	2.5	0.4	2.5	2.3	2.4	2.5	2.1	2.3	2.2	2.3
Growth of real primary spending (deflated by GDP deflator, in percent)	9.6	8.3	7.4	8.8	14.4	12.4	3.9	5.3	5.1	8.3	6.3	6.1	6.1
Grant element of new external borrowing (in percent)	0.0	0.0	0.0	0.5	0.4	0.4	0.4	0.4	0.5	0.4	0.4	0.4	...

Sources: Burkinabè authorities; and IMF staff estimates and projections.

[1] Historical averages and standard deviations are generally derived over the past 10 years, subject to data availability.

[2] Central government

[3] Gross financing need is defined as the primary deficit plus debt service plus the stock of short-term debt at the end of the last period.

[4] Revenues including grants.

[5] Debt service is defined as the sum of interest and amortization of medium and long-term debt.

Appendix Table I.4. Burkina Faso: Sensitivity Analysis for Key Indicators of Public Debt, 2005-25
(In percent)

	Projections							
	2005	2006	2007	2008	2009	2010	2015	2025
NPV of Debt-to-GDP Ratio								
Baseline	22	14	14	16	17	18	21	22
A. Alternative scenarios								
A1. Real GDP growth and primary balance at historical averages	22	14	13	13	14	14	17	24
A2. Primary balance unchanged from 2006	22	14	14	15	17	18	25	37
A3. Permanently lower GDP growth [1]	22	14	14	16	18	19	24	31
B. Bound tests								
B1. Real GDP growth at historical average minus one standard deviations in 2007-08	22	14	15	17	19	20	25	28
B2. Primary balance at historical average minus one standard deviation in 2007-08	22	14	13	14	16	16	20	21
B3. Combination of B1-B2 using one-half standard deviation shocks	22	14	13	14	15	16	19	20
B4. One-time 30 percent real depreciation in 2007	22	14	18	18	18	18	19	19
B5. 10 percent of GDP increase in other debt-creating flows in 2007	22	14	24	25	26	26	26	24
NPV of Debt-to-Revenue Ratio [2]								
Baseline	134	76	79	88	92	94	101	100
A. Alternative scenarios								
A1. Real GDP growth and primary balance at historical averages	134	76	70	72	73	74	83	110
A2. Primary balance unchanged from 2006	134	76	75	83	88	94	122	172
A3. Permanently lower GDP growth 1/	134	76	79	89	94	98	113	140
B. Bound tests								
B1. Real GDP growth at historical average minus one standard deviation in 2007-08	134	76	81	93	99	103	117	127
B2. Primary balance at historical average minus one standard deviation in 2007-08	134	76	72	77	82	85	94	96
B3. Combination of B1-B2 using one-half standard deviation shocks	134	76	71	75	80	83	91	93
B4. One-time 30 percent real depreciation in 2007	134	76	97	99	97	95	91	89
B5. 10 percent of GDP increase in other debt-creating flows in 2007	134	76	134	138	136	135	126	110
Debt Service-to-Revenue Ratio [2]								
Baseline	8.2	6.2	4.9	4.9	4.7	4.4	4.4	5.4
A. Alternative scenarios								
A1. Real GDP growth and primary balance at historical averages	8.2	6.2	4.9	3.9	3.7	3.6	3.4	5.5
A2. Primary balance unchanged from 2006	8.2	6.2	4.9	4.5	4.4	4.4	5.0	8.8
A3. Permanently lower GDP growth [1]	8.2	6.2	4.9	4.9	4.8	4.6	4.8	7.3
B. Bound tests								
B1. Real GDP growth at historical average minus one standard deviations in 2007-08	8.2	6.2	5.0	5.1	5.0	4.8	5.0	6.8
B2. Primary balance at historical average minus one standard deviations in 2007-08	8.2	6.2	4.9	4.2	4.0	4.2	3.9	5.1
B3. Combination of B1-B2 using one-half standard deviation shocks	8.2	6.2	4.9	4.1	3.9	4.1	3.7	4.9
B4. One-time 30 percent real depreciation in 2007	8.2	6.2	5.1	5.4	5.4	5.2	5.3	6.6
B5. 10 percent of GDP increase in other debt-creating flows in 2007	8.2	6.2	4.9	4.9	4.7	4.4	4.4	5.4

Sources: Country authorities; and IMF staff estimates and projections.

[1] Assumes that real GDP growth is at baseline minus one standard deviation divided by the square root of 20 (i.e., the length of the projection period).

[2] Revenues are defined inclusive of grants.